COMPUTACIÓN Y ESTRUCTURAS DISCRETAS III / RESEARCH POSTER

ResumeLens

A formal-language pipeline for résumé screening

Juan Felipe Correa · Juan Diego Garcés · Jhostin Felipe Wiesner

Universidad Icesi · Colombia

METHODS: FOUR FORMAL STAGES, ONE REUSABLE SYSTEM

01 EXTRACT

Regular expressions

Python `re` identifies recognized sections, contact fields and raw skill tokens.

02 NORMALIZE + ORDER

FST + catalog rules

Map aliases to canonical names; filter by profile, remove duplicates and sort by category.

03 RECOGNIZE

Deterministic finite automaton

Accept only when every required category is represented in the prescribed order.

04 VALIDATE + RENDER

Context-free grammar + textX

Serialize a candidate DSL, parse and validate it, then generate HTML.

INTRODUCTION & QUESTION

PROBLEM

Skill names vary (e.g., JS / JavaScript), while profiles require different combinations of skill categories.

OBJECTIVE

Normalize explicit skills and validate them against a selected profile pattern.

RESEARCH QUESTION

Can formal-language models turn varied résumé text into a deterministic, structured profile evaluation?

METHOD CHOICES

REGEX	Extract fields and skill tokens from recognized résumé sections.
FST	Translate aliases and naming variants into canonical symbols.
DFA	Recognize an ordered sequence that meets the selected profile pattern.
CFG / textX	Validate the candidate language before rendering the HTML profile.

IMPLEMENTATION AT A GLANCE

- Python orchestrates a single end-to-end pipeline.
- A shared catalog defines canonical skills, aliases, categories and profile eligibility.
- Streamlit provides profile selection, résumé upload, evaluation status and HTML download.
- Input: UTF-8 plain-text .txt résumé. Output: ACCEPTED / REJECTED plus a validated HTML profile.

RESULTS · DATA ENGINEER EXAMPLE

RAW SKILLS · ORIGINAL ORDER

Git · AWS · Airflow · Spark · PostgreSQL · Python

CANONICAL ORDER AFTER NORMALIZATION + ORDERING

PYTHON → POSTGRESQL → SPARK → AIRFLOW → AWS → GIT

ACCEPTED	All six required categories are represented in order.
REJECTED	Without AIRFLOW, the required WORKFLOW category is missing.

ONE ACCEPTED PATH THROUGH THE DFA

Each transition satisfies the next required category.

G1	G2	G3	G4	G5	G6
PYTHON	POSTGRESQL	SPARK	AIRFLOW	AWS	GIT
LANGUAGE	DATABASE	PROCESSING	WORKFLOW	CLOUD	VCS

$$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_4 \rightarrow q_5 \rightarrow q_6$$

The final state q_6 accepts only after LANGUAGE → DATABASE → DATA_PROCESSING → WORKFLOW → CLOUD → VCS.

FOUR PROFILES · ONE ARCHITECTURE

FULL STACK
LANG → FRONTEND → BACKEND → DATABASE → VCS

ML ENGINEER
LANG → DATABASE → ML_LIBRARY → VCS

DEVOPS
LANG → CONTAINER → CI/CD → IaC → CLOUD → VCS

DATA ENGINEER
LANG → DATABASE → PROCESSING → WORKFLOW → CLOUD → VCS

The catalog supplies profile eligibility; each profile declares its category order.

RESULTS & FINDINGS

- Aliases map to one canonical symbol (e.g., JS → JAVASCRIPT).
- Profile ordering filters out-of-profile skills and removes duplicates.
- The DFA gives a transparent ACCEPTED / REJECTED result.
- The candidate DSL is parsed before HTML rendering; output escapes special characters.

SYSTEM AT A GLANCE

4	39	12
professional profiles	canonical qualifications	skill categories

TEST COVERAGE

157 test cases

94 test functions across 7 modules, including parameterized cases.

SCOPE & LIMITATIONS

- Input is limited to UTF-8 plain-text .txt résumés.
- Skills are extracted only from recognized résumé sections.
- Each profile requires every category in its configured order.
- Unknown skills are not inferred; ResumeLens does not rank candidates or make hiring decisions.

CONCLUSION

Formal models provide an explicit, reproducible path from raw résumé text to a validated candidate profile. Each evaluation can be traced to the catalog and the selected profile rules.

Rule-based evaluation, not an automated hiring decision.

SELECTED REFERENCES

[1] Sipser, M. (2012). Introduction to the Theory of Computation (3rd ed.). Cengage Learning.
[2] Mohri, M. (1997). Finite-State Transducers in Language and Speech Processing. Computational Linguistics, 23(2), 269–311.
[3] pyformlang.readthedocs.io • [4] textx.github.io/textX

PROJECT REPOSITORY github.com/JhostinWiesner/PanchitosFC-ResumeLens

ACKNOWLEDGMENTS

We acknowledge the course guidance and the open-source communities behind pyformlang, textX and Streamlit.

TRY THE LIVE DEMO

Scan to open ResumeLens
Upload a sample résumé

panchitos-fc.streamlit.app